BAUX: Mathematical Specifications, Mechanized Semantics, and BLAKE3 / SLSA v1.0 Compiled-Artifact Provenance

Kvutza
https://github.com/Kvutza/baux · https://kvutza.github.io/baux

November 2026

Abstract

This monograph specifies the mathematical algorithms, mechanized Lean 4.35 theorems, and cryptographic artifact provenance of `baux` (MIT License, Copyright © 2026 Kvutza). Unlike conventional documentation that can drift from compiled binaries or rely on manual license checklists, `baux` couples a two-layer verification architecture: (1) exact mathematical specifications backed by 189 Lean 4.35 modules (3,616 declarations, 1,170 theorems, zero `sorry`, and zero custom axioms) extracted from 4,366 Charon LLBC bodies via Aeneas; and (2) an in-tree parallel **BLAKE3** Merkle engine (`baux::artifact::blake3`) and **in-toto Statement v1 / SLSA Provenance v1.0** attestation pipeline (`baux::artifact::attestation`) that binds source locks, Lean `.olean` objects, Mach-O `arm64` binaries, Apple Silicon Metal kernels, and OxCaml / Hardcaml SIMD circuits while deterministically compiling the repository `NOTICE` file from the resolved dependency graph.

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1 Two-Layer Verification & BLAKE3 / SLSA v1.0 Provenance

Every component of `baux` is governed by a dual-artifact discipline:

1. **Mathematical & Mechanized Layer**: Every optimization, surrogate, pretraining, and hardware kernel is specified in this LaTeX monograph and mechanized in Lean 4.35 under `research/lean/BauxResearch`.

2. **Compiled-Artifact & Attestation Layer**: Every compiled binary, lockfile, and legal attribution is bound by its 256-bit **BLAKE3** digest inside an **in-toto v1 / SLSA Provenance v1.0** statement (`research/extraction/provenance.intoto.json`).

Definition 1.1 (BLAKE3 Binary Merkle Tree Digest [Artifact: `baux::artifact::blake3`])**.** Let $M \in \{0,1\}^*$ be an input byte stream partitioned into $C = \max(1, \lceil |M|/1024 \rceil)$ chunks $M_0, \ldots, M_{C-1}$ of at most 1024 bytes, where each chunk M_c is further divided into $1 \le B_c \le 16$ blocks of at most 64 bytes. Let $\mathrm{IV} \in (\mathbb{Z}/2^{32}\mathbb{Z})^8$ be the standard SHA-256 / BLAKE3 initialization vector and let

$$F_{\mathrm{compress}} : (\mathbb{Z}/2^{32}\mathbb{Z})^8 \times (\mathbb{Z}/2^{32}\mathbb{Z})^{16} \times \mathbb{Z}/2^{64}\mathbb{Z} \times \mathbb{Z}/2^{32}\mathbb{Z} \times \mathbb{Z}/2^{32}\mathbb{Z} \longrightarrow (\mathbb{Z}/2^{32}\mathbb{Z})^{16}$$

denote the 7-round BLAKE3 permutation with message word schedule $\sigma = (2, 6, 3, 10, 7, 0, 4, 13, 1, 11, 12, 5, 9, 14, 15, 8)$. For $C > 1$ chunks, the binary Merkle reduction splits C chaining values into a left complete power-of-two subtree of size $L = 2^{\lfloor \log_2(C-1) \rfloor}$ and a right subtree of size $C - L$:

$$\mathrm{CV}(M_{0..C}) = \pi_{0..7}\Big(F_{\mathrm{compress}}\big(\mathrm{IV},\ \mathrm{CV}(M_{0..L}) \,\|\, \mathrm{CV}(M_{L..C}),\ 0,\ 64,\ \mathrm{PARENT} \cup \mathrm{ROOT}_{\mathrm{top}}\big)\Big).$$

Specification 1.2 (SLSA v1.0 Provenance & Auto-Compiled `NOTICE` [Artifact: `baux::artifact::attestation`])**.** The provenance manifest `research/extraction/provenance.intoto.json` conforms to `https://in-toto.io/Statement/v1` with predicate type `https://slsa.dev/provenance/v1`:

- `subject`: records the exact byte size and `digest.blake3` of every tracked license, lockfile (`research/upstream/*.lock.toml`), Charon LLBC module (`research/extraction/BauxBo.llbc`), OxCaml / Hardcaml source, and this compiled monograph (`papers/baux\_specification.pdf`).
- `predicate.buildDefinition.resolvedDependencies`: enumerates all 17 foundational upstream repositories, clean-room mathematical specifications, and statically linked or vendored libraries alongside their SPDX license identifiers and academic citations.
- **Zero-Drift Attribution**: `verify_license_and_notice` in `baux::artifact::attestation` verifies during `baux-unit` that `NOTICE` and `rust/crates/baux/NOTICE` equal the deterministic projection of `resolvedDependencies` byte-for-byte.

2 Mathematical Specifications of Core Algorithms

2.1 Epistemic Nearest Neighbors (ENN) & Leave-One-Out Fitting

The epistemic nearest-neighbor surrogate in `rust/crates/baux/src/fit.rs` and `rust/crates/baux/semantics/src/fit.rs` ports and formalizes the K-nearest-neighbor epistemic uncertainty estimator of Sweet [1] [Lean: `BauxResearch.Semantics.Neighbor`, `BauxResearch.Semantics.Fit`].

Specification 2.1 (ENN Posterior & Stabilized LOO Log-Likelihood)**.** Given n observed design points $\{(x_i, y_i)\}_{i=1}^n \subset \mathbb{R}^d \times \mathbb{R}$ and length-scale weights $w \in \mathbb{R}^d_{>0}$, define the weighted squared-L_2 distance

$$d_w^2(x, x_j) = \sum_{m=1}^{d} w_m (x_m - x_{j,m})^2.$$

For a query $x \in \mathbb{R}^d$ and its K nearest neighbors $\mathcal{N}_K(x) \subset \{1, \ldots, n\}$, let $r_j = \max\bigl(\sqrt{d_w^2(x, x_j)}, \varepsilon_0\bigr)$ with inverse-distance weights $\alpha_j = r_j^{-2}$ and normalized weights $\tilde{\alpha}_j = \alpha_j / \sum_{k \in \mathcal{N}_K(x)} \alpha_k$. The predictive mean $\mu(x)$ and epistemic standard error $\sigma(x)$ are

$$\mu(x) = \sum_{j \in \mathcal{N}_K(x)} \tilde{\alpha}_j y_j, \qquad \sigma(x) = \sqrt{\frac{\hat{v}(x) + \sigma_{\text{noise}}^2}{\sum_{j \in \mathcal{N}_K(x)} \alpha_j + \kappa_0}}.$$

During leave-one-out (LOO) cross-validation at held-out index $i \in \{1, \ldots, n\}$:

1. Self-exclusion is enforced by exact index inequality $j \neq i$ rather than a distance cutoff $d_w^2(x_i, x_j) > 10^{-15}$, preventing duplicate observations from corrupting LOO estimates.
2. The Gaussian log-likelihood is evaluated in log-standard-deviation form:

$$\ell_i(\mu_{-i}, \sigma_{-i}) = -\frac{1}{2}\ln(2\pi) - \ln(\sigma_{-i}) - \frac{1}{2}\left(\frac{y_i - \mu_{-i}}{\sigma_{-i}}\right)^2,$$

   avoiding IEEE-754 `binary64` underflow of σ_{-i}^2 when $\sigma_{-i} < 10^{-154}$.

2.2 Trust Region Bayesian Optimization (TuRBO) & Multi-Objective MORBO

The regional trust-region controller in `rust/crates/baux/src/trust\_region.rs` and `rust/crates/baux/src/morbo.rs` implements the clean-room mathematical specifications of Eriksson et al. [2] and Daulton et al. [3] [Lean: `BauxResearch.Semantics.TrustRegion`, `BauxResearch.Semantics.Hypervolume`].

Specification 2.2 (TuRBO State Machine & Sobol Perturbation Mask)**.** A trust region over $[0,1]^d$ maintains base edge length $L \in [L_{\min}, L_{\max}]$ with defaults $L_{\text{init}} = 0.8$, $L_{\min} = 2^{-7}$, $L_{\max} = 1.6$, success tolerance $\tau_{\text{succ}} = 3$, and failure tolerance $\tau_{\text{fail}} = \max\bigl(4, \lceil d/q \rceil\bigr)$ for batch size $q \geq 1$. After each batch evaluation:

$$(c'_{\text{succ}}, c'_{\text{fail}}) = \begin{cases} (c_{\text{succ}} + 1,\ 0) & \text{if } \min_{b=1..q} f(x_b) < f(x^*) - 10^{-3}|f(x^*)|, \\ (0,\ c_{\text{fail}} + 1) & \text{otherwise,} \end{cases}$$

and the side length updates via

$$L' = \begin{cases} \min(2L,\ L_{\max}) & \text{if } c'_{\text{succ}} = \tau_{\text{succ}}, \\ L/2 & \text{if } c'_{\text{fail}} = \tau_{\text{fail}}, \\ L & \text{otherwise,} \end{cases}$$

triggering a restart when $L' < L_{\min}$. Candidate generation perturbs the incumbent x^* inside $\prod_{m=1}^{d} \bigl[\max(0, x_m^* - \tfrac{1}{2}\lambda_m L),\ \min(1, x_m^* + \tfrac{1}{2}\lambda_m L)\bigr]$ Along active coordinates sampled with probability $p_{\text{mask}} = \min(1, 20/d)$ (requiring at least one active coordinate per candidate).

2.3 Polar Express & Muon Orthogonalized Momentum

The matrix-sign preconditioning and orthogonalized momentum optimizer in `rust/crates/baux/src/muon/` and `rust/crates/baux/semantics/src/muon.rs` formalize Amsel et al. [6] and Jordan et al. [5] [Lean: `BauxResearch.Semantics.Muon`].

Specification 2.3 (Polar Express Minimax Odd Quintic & Newton-Schulz Iteration)**.** Given a momentum matrix $G \in \mathbb{R}^{m \times n}$, define the Frobenius-normalized iterate $X_0 = G/(\|G\|_F \cdot 1.01 + \varepsilon)$ (reducing transposed when $m > n$). At iteration $t = 0, \ldots, T-1$, let $A_t = X_t X_t^\top$ (or $X_t^\top X_t$ when $m > n$) and apply the odd quintic polynomial

$$X_{t+1} = a_t X_t + \bigl(b_t A_t + c_t A_t^2\bigr) X_t.$$

- **Muon 5-step mode**: fixes $(a_t, b_t, c_t) = (3.4445, -4.7750, 2.0315)$ for all $t \in \{0, \ldots, 4\}$.
- **Polar Express minimax mode**: evolves the spectral interval $[\ell_t, u_t]$ from $[\ell_0, 1]$ with cushion $\tilde{\ell}_t = \max(\ell_t, u_t/10)$ via Remez equioscillation so that $\max_{\sigma \in [\tilde{\ell}_t, u_t]} |1 - (a_t\sigma + b_t\sigma^3 + c_t\sigma^5)|$ is minimized at each step, followed by safety deflation $a_t \leftarrow a_t/1.01$, $b_t \leftarrow b_t/1.01^3$, $c_t \leftarrow c_t/1.01^5$.

2.4 Dynamic Chunking (H-Net) & Manifold-Constrained Hyper-Connections (mHC)

The hierarchical sequence chunking and doubly stochastic residual mixing modules in `rust/crates/baux/semantics/src/hnet/` and `rust/crates/baux/src/mhc.rs` formalize Hwang, Wang, and Gu [4] and Xie et al. [7] [Lean: `BauxResearch.Semantics.HNet`, `BauxResearch.Semantics.Mhc`].

Specification 2.4 (H-Net Cosine Boundary Routing & EMA Dechunk Scan)**.** Given projected query and key streams $q_t, k_t \in \mathbb{R}^D$ for token positions $t = 0, \ldots, L-1$:

$$p_0 = 1, \qquad p_t = \frac{1}{2}\left(1 - \frac{\langle q_t, k_{t-1}\rangle}{\max(\|q_t\|_2 \, \|k_{t-1}\|_2 \, , \, \varepsilon)}\right) \in [0, 1] \quad (t \geq 1),$$

with discrete boundary indicator $b_t = \mathbf{1}[p_t \geq 1/2]$ and straight-through confidence $c_t = b_t p_t + (1 - b_t)(1 - p_t)$. Given compressed chunk representations $\hat{z}_k = c_k z_k$, the dechunking operator computes the causal exponential moving average

$$h_0 = p_0 \hat{z}_0, \qquad h_t = (1 - p_t) h_{t-1} + p_t \hat{z}_{\mathrm{idx}(t)},$$

executed on Apple Silicon Metal via a two-pass affine parallel prefix scan in `rust/crates/baux/src/hnet\_wave.metal`.

Specification 2.5 (mHC Sinkhorn-Knopp Birkhoff Polytope Projection)**.** Given raw stream-mixing logits $W \in \mathbb{R}^{S \times S}$ across S residual streams, let $P^{(0)}_{i,j} = \exp(W_{i,j} - \max_k W_{i,k})$. For iterations $r = 1, \ldots, R$, alternate row and column normalizations

$$\widetilde{P}^{(r)}_{i,j} = \frac{P^{(r-1)}_{i,j}}{\sum_{k=1}^{S} P^{(r-1)}_{i,k} + \varepsilon}, \qquad P^{(r)}_{i,j} = \frac{\widetilde{P}^{(r)}_{i,j}}{\sum_{i'=1}^{S} \widetilde{P}^{(r)}_{i',j} + \varepsilon},$$

yielding a doubly stochastic matrix $P^{(R)} \in \mathcal{B}_S$ that preserves the convex-hull norm invariant $\left\|P^{(R)} X\right\|_{\mathrm{op}} \leq \|X\|_{\mathrm{op}}$.

2.5 OxCaml Unboxed SIMD (`float64x2#`) & Hardcaml RTL Distance Circuits

The native numerical benchmark and hardware synthesis suite in `ocaml/` compiles 56 Simon Fraser University (SFU) global optimization test functions under OxCaml 5.2.0minus-ox using unboxed `float#` scalars, ARM64 NEON `float64x2#` 128-bit vector intrinsics, and SLEEF 3.9.0 vector transcendentals, alongside a pipelined Hardcaml register-transfer-level (RTL) circuit for squared-L_2 distance accumulation.

Specification 2.6 (OxCaml Unboxed SIMD & Hardcaml Distance Datapath)**.** 1. **Zero-Allocation Unboxed Evaluation** (`ocaml/lib/formulas.ml`): Every 2-lane vector kernel evaluates coordinate pairs in unboxed `float64x2#` registers (`fadd.2d`, `fmul.2d`, `fmla.2d`) with SLEEF 3.9.0 1-ULP vector transcendentals (`Sleef_cosd2`, `Sleef_sind2`, `Sleef_expd2`), verified against SciPy reference values to relative tolerance $\leq 10^{-12}$.

2. **Hardcaml RTL Distance Circuit (`ocaml/hardware/distance.ml`)**: A 3-stage synchronous pipeline over 64-bit IEEE-754 inputs $(x_{\text{bits}}, y_{\text{bits}})$ computes Stage 1 difference $\delta_t = x_t - y_t$, Stage 2 square $s_t = \delta_t^2$, and Stage 3 accumulation $A_t = \mathbf{1}[\neg\text{clear}_t]A_{t-1} + \mathbf{1}[\text{valid}_t]s_t$, matching software `Float.fma` bit-for-bit across all cycle-accurate testbenches.

3 Compiled-Artifact Correspondence & Symbol Audit

Table 1 records the compiled binary and proof artifacts tracked by the `baux` verification pipeline, the build toolchains that produce them, and the symbol and digest predicates enforced by `./baux check` and `research/extraction/provenance.intoto.json`.

Artifact / Binary	Format / Target	Toolchain	Verification Predicate
`libbaux.rlib`	`ar+mach-o-64 arm64`	Buck2 + `rustc`	BLAKE3 + `nm -g` C-ABI symbols
`baux_rust.so`	`mach-o-64 arm64`	Buck2 + PyO3	BLAKE3 + Python FFI parity
`libbaux_numerical.so`	`mach-o-64 arm64`	Lean 4.35 C-ABI	BLAKE3 + 7-word IEEE-754 check
`BauxBo.llbc`	Charon LLBC v0.1.178	Charon MIR	BLAKE3 + 4,366 structured bodies
`BauxResearch.olean`	Lean 4.35 `.olean`	`lake build`	189 modules, 0 `sorry`, 0 custom axioms
`hnet_wave.metal`	Apple Metal Shading	Metal / MPS	2-pass affine scan vs. CPU bit-parity
`benchmarks.exe`	`mach-o-64 arm64`	OxCaml + SLEEF	`otool -tvV .2d` NEON + 0 `caml_alloc`
`distance.ml` RTL	Hardcaml RTL netlist	Hardcaml v0.18	Cycle-accurate IEEE-754 testbench
`provenance.intoto.json`	in-toto v1 / SLSA v1.0	`baux::artifact`	BLAKE3 subject & `NOTICE` bijection

Table 1: Compiled artifacts, native binaries, and formal verification targets in `baux`.

4 Open-Source License & Academic Provenance Inventory

Table 2 lists the complete open-source and academic provenance inventory compiled into `NOTICE`, `rust/crates/baux/NOTICE`, and `research/extraction/provenance.intoto.json`. All non-commercial upstream locks have been excluded from the repository.

References

[1] D. Sweet, "Tsallis Entropy Guided By Epistemic Nearest Neighbors," *arXiv preprint arXiv:2502.10390*, 2025.

[2] D. Eriksson, M. Pearce, J. R. Gardner, R. Turner, and M. Poloczek, "Scalable Global Optimization via Local Bayesian Optimization," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019, *arXiv:1910.01739*.

[3] S. Daulton, D. Eriksson, M. Balandat, and E. Bakshy, "Multi-Objective Bayesian Optimization over High-Dimensional Search Spaces," in *Uncertainty in Artificial Intelligence (UAI)*, 2022, *arXiv:2109.10964*.

[4] S. Hwang, B. Wang, and A. Gu, "Dynamic Chunking for End-to-End Hierarchical Sequence Modeling," *arXiv preprint arXiv:2507.07955*, 2025.

Component	SPDX License	Copyright / Upstream	Citation / Role in baux
baux (this work)	MIT	© 2026 Kvutza	Core Rust, Lean 4, Metal, OxCaml, Python
yubo-research/enn	MIT	© 2025 yubo research	Sweet [1]; ENN foundation
TuRBO / BoTorch	MIT	Eriksson et al.; Meta Platforms	Eriksson et al. [2]; clean-room spec
MORBO	MIT	Daulton et al.; Meta Platforms	Daulton et al. [3]; clean-room spec
goombalab/hnet	Apache-2.0	© 2025 Goomba Lab	Hwang et al. [4]; dynamic chunking
Apple mlx	MIT	© 2023–2024 Apple Inc.	Metal SIMD scan idioms (hnet_wave.metal)
KellerJordan/Muon	MIT	© 2024 Keller Jordan	Jordan et al. [5]; Muon update
PolarExpress	MIT	© 2025 Amsel et al.	Amsel et al. [6]; matrix sign
mHC	MIT	© 2025 tokenbender / Xie et al.	Xie et al. [7]; Sinkhorn-Knopp
DeepMind torax	Apache-2.0	© 2024 DeepMind Technologies	Citrin et al. [8]; tokamak PDE
TokaGym	MIT	© 2025 Wang, Theiler et al.	Wang et al. [9]; TCV rampdown
DeepMind acme	Apache-2.0	© 2018 DeepMind Technologies	Hoffman et al. [10]; RL loop
OxCaml & Hardcaml	MIT / LGPL-2.1*	© Jane Street Group, LLC	Unboxed float64x2# & RTL circuits
SLEEF v3.9.0	BSL-1.0	© Naoki Shibata	ARM64 NEON vector transcendentals
SciPy Optimize	BSD-3-Clause	© Enthought / SciPy Devs	56 SFU global optimization test vectors
Unum usearch	Apache-2.0	© 2023 Ash Vardanian / Unum	Vendored HNSW vector index (vendor/)
Candle & PyO3	Apache-2.0/MIT	© Hugging Face / PyO3 Devs	Tensor backend & Python FFI bindings

Table 2: Complete software, mathematical specification, and third-party license provenance.

[5] K. Jordan et al., "Muon: An optimizer for hidden layers in neural networks," https://github.com/KellerJordan/Muon, 2024.

[6] N. Amsel, D. Persson, C. Musco, and R. M. Gower, "The Polar Express: Optimal Matrix Sign Methods and Their Application to the Muon Algorithm," *arXiv preprint arXiv:2505.16932*, 2025.

[7] Z. Xie, Y. Wei, H. Cao, et al., "mHC: Manifold-Constrained Hyper-Connections," *arXiv preprint arXiv:2512.24880*, 2025.

[8] J. Citrin et al., "TORAX: A Fast and Differentiable Tokamak Transport Simulator in JAX," *arXiv preprint arXiv:2406.06718*, 2024.

[9] A. M. Wang et al., "Learning plasma dynamics and robust rampdown trajectories with predict-first experiments at TCV," *Nature Communications*, vol. 16, p. 8877, 2025.

[10] M. Hoffman et al., "Acme: A Research Framework for Distributed Reinforcement Learning," *arXiv preprint arXiv:2006.00979*, 2020.